

IBL12307: WEB DEVELOPMENT LABARATORY PRACTICAL

Final Project

PHP & MySQL Web Application Development Project

Total Marks: 100

Submission: Source Code + Database + Documentation + Demonstration

Submit to: edwin.ngwawe@tukenya.ac.ke

Objective

Develop a complete PHP-MySQL web application of your own choice that demonstrates the knowledge and skills acquired throughout this course.

The application should solve a real-world problem and must not be a copy of the Student Registration System developed during the laboratory sessions.

Students are expected to analyse the problem, design the database, implement the system, test it, and present it.

Project Requirements

Choose ONE application from the examples below or propose another application for approval.

Examples include:

- Library Management System
- Inventory Management System
- Employee Management System
- Clinic Appointment System
- Hospital Patient Records System
- Hotel Booking System
- Pharmacy Management System
- School Fees Management System
- Vehicle Service Management System
- Event Registration System
- Church Membership System
- Restaurant Ordering System
- Hostel Management System

- Customer Relationship Management System
- Online Course Registration System

Students may propose another system provided it demonstrates similar complexity.

Functional Requirements

Your system should include the following features.

1. User Authentication (10 Marks)

The application must allow users to:

- Register
- Login
- Logout
- Change Password

Passwords must be securely hashed.

2. User Roles (10 Marks)

Implement at least two user roles.

Example:

Administrator

Can:

- Add
- Edit
- Delete

Normal User

Can:

- View only

Restrict pages appropriately.

3. Database Design (10 Marks)

Design a relational database.

Requirements

- Minimum four related tables
- Primary keys
- Foreign keys where appropriate
- Proper normalization

Students should provide an ER Diagram.

4. CRUD Operations (15 Marks)

Implement Create

Read

Update

Delete

for the main records of the system.

5. Search and Pagination (10 Marks)

Users should be able to:

- Search records
- Navigate large datasets using pagination

6. File Upload (10 Marks)

Upload at least one type of file.

Examples

- Passport photo
- Product image
- Employee photo
- Book cover

Validate:

- File type
- File size

7. Validation (10 Marks)

Implement

Client-side validation

Using JavaScript.

Server-side validation

Using PHP.

Validate:

- Required fields
- Email
- Duplicate records
- Invalid input

8. Security (10 Marks)

The application should demonstrate:

- Prepared Statements
- SQL Injection protection
- XSS prevention
- Password hashing
- Session protection

9. Activity Logging (5 Marks)

Maintain an activity log showing:

- Login
- Logout
- Add
- Edit
- Delete

Include

- User

- Activity
- Date
- Time

10. User Interface (5 Marks)

Your application should have:

- Professional layout
- Consistent navigation
- CSS or Bootstrap styling
- Meaningful messages

11. Testing (5 Marks)

Test the application.

Include evidence that you tested:

- Login
- CRUD
- Validation
- Security
- Search
- File upload

Document any bugs found and how they were resolved.

Documentation Requirements

Each student should prepare a report containing:

Chapter 1

Introduction

- Problem Statement
- Objectives
- Scope

Chapter 2

System Analysis

- Functional Requirements
- Non-functional Requirements

Chapter 3

System Design

Include

- ER Diagram
- Database Tables
- Use Case Diagram
- System Flowchart

Chapter 4

Implementation

Explain:

- Technologies Used
- Database
- Major Modules
- Screenshots

Chapter 5

Testing

Include:

- Test cases
- Expected results
- Actual results

Chapter 6

Conclusion

- Challenges
- Lessons Learned
- Recommendations
- Future Improvements

Submission Requirements

Each student should submit:

1. Complete project folder
2. SQL database export (.sql)
3. Project report (PDF)
4. PowerPoint Presentation
5. Screenshots of the system
6. Source code

Presentation

Each group will give a **10–15 minute presentation** demonstrating:

- The problem being solved
- Database design
- Major modules
- Live demonstration
- Challenges encountered
- Lessons learned

Students should be prepared to answer questions about their code.

Assessment Rubric (100 Marks)

Component	Marks
Problem Selection & Originality	5
Database Design	10
Authentication	10
User Roles & Authorization	10
CRUD Operations	10
Search & Pagination	5
File Upload	5
Validation & Security	5
Activity Logging	5
User Interface	5
Documentation	5
Presentation & Demonstration	25
Total	100